ALI GARABAGLU

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RESEARCH INTERESTS

Interested in experimental particle physics, with a focus on applying data science and machine learning methods to advance research.

EDUCATION

University of Washington, Seattle, WA **2021-present**
PhD candidate in Physics
- **Advisor:** Quentin Buat

Rutgers University, New Brunswick, NJ **2016-2020**
B.S. magna cum laude in Physics with High Honors, Minor in Mathematics
- **Thesis:** “A Study of the MATHUSLA Test Stand and Proposed Large Scale Geometries”
- **Advisor:** John Paul Chou

SELECTED PAPERS

ATLAS:
- A. Garabaglu, Q. Buat and A. De Maria, “Identification of highly collimated hadronically decaying τ leptons”, tech. rep., 2024, url: https://cds.cern.ch/record/2906623

FASER:
- FASER Collaboration, “First Measurement of the ν_e and ν_μ Interaction Cross Sections at the LHC with FASER's Emulsion Detector”, arXiv preprint arXiv:2403.12520 (2024).

MATHUSLA:
- Alpigiani Cristiano et al., "An Update to the Letter of Intent for MATHUSLA: Search for Long-Lived Particles at the HL-LHC." arXiv preprint arXiv:2009.01693 (2020).
- Alidra, Maf, et al. "The MATHUSLA test stand." Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment 985 (2021): 164661
- Alpigiani Cristiano et al., "Recent Progress and Next Steps for the MATHUSLA LLP Detector". arXiv preprint arXiv:2203.08126 (2022).

CMS notes:
- The CMS Collaboration. "Search for physics beyond the standard model in high-mass diphoton events from proton-proton collisions at $\sqrt{s}$ = 13 TeV." Physical Review D 98.9 (2018): 092001.
- The CMS Collaboration. "Search for Pair-Produced Resonances Each Decaying into at Least Four Quarks in Proton-Proton Collisions at $\sqrt{s}$ = 13 TeV." Physical review letters 121.14 (2018): 141802.

RESEARCH EXPERIENCE

- **Research Assistant, University of Washington** **2021-present**

Worked on data extraction and analysis from the FASER emulsion detector during the first two years, before transitioning to the ATLAS experiment where I focused on developing di-tau identification and reconstruction algorithms and leading the boosted Higgs to di-tau analysis.

- **Research Assistant, Rutgers University** **2017-2021**

Developed and implemented fitting procedures, Monte Carlo production workflows, and software frameworks for two CMS analyses, and contributed to the design and development of the GEANT4 simulation framework for the MATHUSLA detector.

TEACHING

- Teaching assistant for High Energy Physics Winter 2023
- Teaching assistant for Neural Network Methods for Signals in Engineering and Physical Sciences Spring 2022(23)
- Teaching assistant for Digital Electronics Lab Spring 2022
- Teaching assistant for Mathematical Physics Winter 2022
- Teaching assistant for Computational Data Analysis Lab Autumn 2022
- Teaching assistant for Nuclear & Particle Physics Experiments Lab Autumn 2021

TALKS

- Probing the High-Momentum Frontier of Higgs Boson Decay to a pair of Tau Leptons, 2024, ATLAS US Workshop, Seattle, WA.
- Boosted DiTau High + Low pt Reconstruction and Identification, 2024, TauCP + HLepton Workshop, Geneva, Switzerland.
- Results From TeV Neutrinos at the FASER Experiment, 2024, DPF-PHENO, Pittsburg, PA.
- FASER Tracking and Emulsion Station Alignment, 2024, ACAT Conference, Stony Brook, NY.
- Weakly-Supervised Anomaly Detection with Conditional VAEs, 2023, APS, online.

AWARDS

- University of Washington Provost Award 2021
- Rutgers University SAS Paul Robeson Scholar 2020
- Rutgers University Robert L. Sells Scholarship 2020
- Rutgers University SAS Excellence Award 2019
- Rutgers University Jaqua Scholarship 2019